

## MA 322: Applied Graph Theory

Course Code: MA 322

Course Name: Applied Graph Theory

L-T-P-C: 3-0-2-4

Prerequisites:

Students intended for: B.Tech. Mathematics & Computing

Elective or Core: Core

Mutual Exclusion: MA 530

Approval: 59th BoA

### Course Contents

- **Fundamentals of Graph Theory:** Basic definitions of graphs, adjacency, degree, degree sequence, graph isomorphism, walks, trails, path, cycles, bipartite graphs and their characterizations, subgraphs, graph representations: adjacency matrix and adjacency list, directed graphs, tournaments. [4 Lectures]
- **Trees and Connectivity in Graphs:** Definition and properties of trees, forests, spanning trees, Cayley's formula, Prim's and Kruskal's algorithm, Dijkstra's Algorithm and Bellman-Ford algorithm, vertex cut, edge cut, applications of cuts, Menger's theorem. [10 Lectures]
- **Graph Traversal and Shortest Paths:** Breadth-First Search (BFS), Depth-First Search (DFS), topological sorting and directed acyclic graphs, strongly connected components in directed graphs, All-Pairs Shortest Paths: Floyd-Warshall Algorithm. [5 Lectures]
- **Vertex Cover Independent Sets Flows and Matching:** Vertex covers and independent sets in a graph, matching, matching in bipartite graphs, Hall's Marriage theorem, Stable Matching and the Gale-Shapley Algorithm, Network flow, Ford Fulkerson algorithm, residual graphs, and augmenting paths. [12 Lectures]
- **Eulerian and Hamiltonian Graphs, Coloring and Planarity:** Eulerian graphs, Fleury's algorithms, Hamiltonian paths, Hamiltonian cycles and their properties, vertex coloring, applications of vertex coloring, edge coloring, planar graphs and their properties, coloring of planar graphs. [6 Lectures]
- **Spectral Graph Theory and Networks:** Basic definitions and applications of spectral graph theory, real-world networks, including social networks, biological networks, and technological networks. It covers centrality measures (degree, closeness, betweenness, eigenvector centrality, Page Rank), community detection (Girvan-Newman, Louvain method), and random graph models such as Erdos-Renyi and Barabasi-Albert models. [5 Lectures]

### Laboratory/practical/tutorial Modules:

- The algorithms and concepts discussed in the class will be implemented in the lab sessions

### Text books:

1. D. B. West, **Introduction to Graph Theory**, Pearson Education.
2. E. Tardos and J. Kleinberg, **Algorithm Design**, Pearson Education.



#### 4. References:

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, **Introduction to Algorithms**, 3rd Edition, MIT Press.
2. R. Balakrishnan and K. Ranganathan, **A Textbook of Graph Theory**, Springer.
3. R. Diestel, **Graph Theory**, Springer.